Lecture plan · for the lecturer

Teaching modelling and assessing it

Optimization · University · Lecture 4

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Teach the modelling cycle.
- Use open briefs.
- Assess the model.

Activity: Interview, formulate, defend

1. Warm-up (5 min): Show the 30% saving and the transport office's rejection; ask students to guess why.
2. Interview (15 min): Teams question you as the transport officer.
3. Formulate (20 min): Write variables, objective, constraints and assumptions.
4. Defend (15 min): Each team defends its model; others look for a missing requirement.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which teams found the night buses?
- Which assumption was riskiest?
- How will you weight the defence?

Exit ticket

- Write one open brief for your course in three sentences.

Try it in your next lecture
Read a four-sentence brief about the campus shuttle and play the transport officer; teams interview you for ten minutes and formulate, and you reveal the night-bus requirement only to teams who ask.

Seminar sheet · for students

Teaching modelling and assessing it

Optimization · University · Lecture 4

Name _____ Group _____ Date _____

1 **Two teams report the same optimal cost. What best shows who modelled well?**

- ◯ Assumptions, validation and an oral defence
- ◯ The solver they used
- ◯ The length of the report

2 **Add the constraint.**

To guarantee service, add buses[r, t] ≥ _____ whenever demand[r, t] > 0, and require buses[r, t] to be _____ .

Word bank: 0 · 1 · integer · continuous

3 **Which belong in a strong modelling-assignment rubric? Choose all that apply.**

- ☐ Validation against the brief
- ☐ Stated assumptions
- ☐ Only the optimal value
- ☐ Sensitivity analysis

4 **If the solver reports an optimal solution, the plan meets the stakeholders' needs.**

- ◯ True
- ◯ False

5 **Put the modelling cycle in order.**

- ☐ Solve
- ☐ Validate and revise
- ☐ Understand the problem with stakeholders
- ☐ Formulate the model

Answers · for the lecturer

Teaching modelling and assessing it

Optimization · University · Lecture 4

1 **Assumptions, validation and an oral defence**

The model and its justification, not the number.

2 **To guarantee service, add buses[r, t] ≥ 1 whenever demand[r, t] > 0, and require buses[r, t] to be integer.**

At least one bus where there is demand, and whole buses only.

3 **Stated assumptions; Validation against the brief; Sensitivity analysis**

The number alone hides the model's quality.

4 **False**

It is optimal for the model; validation decides whether the model fits reality.

5 **Top to bottom: 3, 4, 1, 2**

Validation sends you back to the start when needed.

For you to think about

Where in your course do students formulate a model from a brief that does not name the variables?